Mathematics

Name:

TEACHER

P.Mac Sweeney

2025/2026

S5EN….

FIRST
SEMESTER

école
internationale
gaston thorn

1. Solve the following using three Methods (factors, formula, complete square)

$x^2 + 6x - 7=0$	$x^2 + 6x - 7=0$	$x^2 + 6x - 7=0$

(8 marks)

2. Solve the following using three Methods (factors, formula, complete square)

$-x^2 + 4x + 5 = 0$	$-x^2 + 4x + 5 = 0$	$-x^2 + 4x + 5 = 0$

(8 marks)

3. Given the function $f(x) = x^2 + 6x - 7$

i. Determine the point where f(x) crosses the y-axis.

ii. Determine the points where f(x) crosses the x-axis

iii. Complete the square to determine the coordinates of the minimum point of the curve

iv. Sketch the graph

(8 marks)

4. Given the function $f(x) = -x^2 - 6x + 7$

ii. Determine the point where f(x) crosses the y-axis.

iii. Determine the points where f(x) crosses the x-axis

iv. Complete the square to determine the coordinates of the maximum point of the curve

v. Sketch the graph

(8 marks)

5. Solve the system of simultaneous equations for x and y

$$x - y = 2$$
$$x^2 + y^2 = 20$$

(8 marks)

6. Matheus is a professional deep water free diver.

His depth (in meters relative to sea level), after x seconds after diving, is modeled by:

$$d(x) = \frac{1}{2}x^2 - 10x$$

i) How many seconds after diving will Rui reach his lowest altitude?

ii) How many seconds after diving will Rui reach the surface again

(5 marks)